Resilio: A Digital Twin-Driven Framework for Disaster Communication and Emergency Response

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***Abstract*—Natural disasters such as floods, earthquakes, cyclones, and landslides frequently damage communication infrastructure, causing cellular networks and internet services to fail. The resulting loss of communication delays emergency response, hampers rescue coordination, and reduces situational awareness, worsening outcomes for affected populations. Existing infrastructure-less communication systems primarily focus on delivering messages between devices and provide little support for emergency decision-making. This paper introduces Resilio, a Digital Twin-driven framework for disaster communication and emergency response that adds real-time situational awareness and intelligent coordination to infrastructure-less networks. The framework integrates device-to-device communication, delay-tolerant message forwarding, a Digital Twin, and an Emergency Decision Support System (EDSS). A multi-gateway architecture collects emergency reports from disaster-affected regions and forwards them to the Digital Twin, which maintains a real-time representation of communication status, disaster conditions, shelters, hospitals, and available rescue resources. The EDSS analyzes this data to prioritize emergencies, recommend resource allocation, and generate response instructions for affected citizens. In doing so, Resilio transforms infrastructure-less disaster communication from a simple message-delivery mechanism into an intelligent emergency response system. The framework is evaluated through a Monte Carlo network simulation using communication reliability metrics (packet delivery ratio, end-to-end delay, and network coverage) and emergency response effectiveness (response time).**

***Index Terms*—Digital Twin, Disaster Communication, Emergency Response, Device-to-Device Communication, Delay-Tolerant Networking, Emergency Decision Support System, Mesh Networking, Disaster Management.**

I. Introduction

Natural disasters such as floods, earthquakes, cyclones, and landslides are occurring with increasing frequency [2], [11]. These events cause severe damage to communication systems and disrupt emergency response capabilities. When cellular networks, internet connectivity, and power infrastructure fail, rescue coordination is delayed and disaster-affected individuals are unable to communicate with emergency responders [2], [11]. Ensuring reliable communication during disasters therefore remains a major challenge for researchers and industry. To address this problem, researchers have investigated device-to-device (D2D) communication, multi-hop mesh networking, and Delay-Tolerant Networking (DTN) as ways for nearby devices to exchange emergency messages without relying on centralized infrastructure [1], [2], [12], [13], [16], [17].

Despite this progress, most existing solutions focus narrowly on message routing and resource efficiency [1], [2]. For instance, prior work has proposed frameworks for scheduling, routing, and device pairing to improve communication efficiency during disasters [1], while other studies have surveyed post-disaster communication technologies and identified several open challenges [2], [11]. In parallel, Digital Twin technology has emerged as a promising approach for creating a real-time virtual representation of a physical environment that supports monitoring, simulation, and decision-making [3]–[6], [24]. Several studies have demonstrated the value of Digital Twins for communication networks and emergency planning [3], [4], [5], [6], [14]. However, most existing work addresses either communication network modeling or disaster visualization in isolation, without integrating communication, resource coordination, and operational decision support into a single framework [3]–[6], [9], [10].

To address this gap, we propose **Resilio**, a Digital Twin-driven framework for disaster communication and emergency response. Resilio extends infrastructure-less disaster communication by integrating multi-hop D2D communication, DTN, a Digital Twin, an Emergency Decision Support System, and a multi-gateway communication architecture [1], [3], [4], [5], [6]. Emergency reports generated by disaster-affected individuals propagate through the mesh communication layer [1], [12], [13] and synchronize with the Digital Twin to maintain a real-time representation of the disaster environment [3], [4], [14]. The EDSS then analyzes these reports to support rescue prioritization and resource allocation [4], [9], [10], and the resulting response instructions are relayed back to affected individuals through the same infrastructure-less network [1], [4].

The main contributions of this paper are: (1) the development of Resilio, a Digital Twin-driven framework that integrates infrastructure-less disaster communication with real-time disaster visualization [1], [3], [14]; (2) a multi-gateway communication architecture for collecting emergency reports from disaster-affected regions [1], [12], [13]; (3) an Emergency Decision Support System that prioritizes rescue requests, coordinates emergency resources, and supports informed decision-making [4], [9], [10]; and (4) a unified evaluation of the framework through simulation using communication reliability and emergency response performance metrics [1], [3],

[4]. The remainder of this paper is organized as follows. Section II reviews related literature. Section III presents the problem statement. Section IV describes the proposed Resilio framework. Section V presents the performance evaluation and discussion. Finally, Section VI concludes the paper and outlines future work.

II. Literature Survey

Recent advances in disaster management have focused on improving communication resilience, situational awareness, and intelligent emergency response [2], [4], [11]. Prior work can be broadly grouped into three areas: infrastructure-less disaster communication, Digital Twin-based disaster management, and Emergency Decision Support Systems [2], [3], [4], [11]. While each area has independently made progress, integrating these technologies into a single disaster response framework remains a challenge [2], [4], [14].

A. Infrastructure-Less Disaster Communication

Several recent studies have examined infrastructure-less disaster communication, including device-to-device communication, mesh networking, and Delay-Tolerant Networking [1], [2], [12], [13]. Peer et al. proposed a framework to improve device-to-device communication and pairing with surviving base stations [1]. Matracia et al. surveyed the technologies used for post-disaster communication and identified Delay-Tolerant Networking as critical for disaster-resilient networking [2]. Debnath et al. discussed the challenges of large-scale disaster response under resource constraints [11]. Other recent works have explored mesh- and IoT-enabled emergency communication [12], [13].

At the protocol level, the DTN architecture was introduced to support communication across intermittently connected and challenged networks [16], [17]. Classical DTN routing schemes such as epidemic routing [18], Spray-and-Wait [19], and PRoPHET [20] address store-carry-forward delivery when end-to-end paths are unavailable. However, these approaches, together with the disaster-specific solutions above, focus primarily on reliable message delivery and provide limited support for emergency coordination and decision-making [1], [2], [11].

B. Digital Twin-Based Disaster Management

Digital Twin technology creates a continuously updated virtual representation of the physical world [3]–[6], [14], [24]. Meuser et al. (NetSkylines) used Digital Twins to evaluate disaster communication systems [3]. Almasan et al. applied Digital Twins to monitor and optimize communication networks [5], [6]. Fan et al. used Digital Twins for city-scale disaster management [14]. Other works have applied Digital Twins and edge intelligence to emergency communication and routing [7], [8]. These studies demonstrate that Digital Twins are effective for monitoring and simulation but are rarely integrated with emergency communication and decision support [3]–[6], [9].

C. Emergency Decision Support Systems

Emergency Decision Support Systems (EDSS) help authorities understand evolving disaster conditions, prioritize rescue operations, and allocate resources [4], [9], [10]. Existing reviews identify Digital Twin-supported decision-making as a promising direction [4], [15], and edge computing and artificial intelligence have been used to process information and coordinate resources [10]. However, most EDSS implementations assume that communication infrastructure remains operational, which limits their applicability when cellular and internet connectivity are unavailable [2], [4], [11].

D. Summary

Existing work has largely treated disaster communication, Digital Twin technology, and emergency decision support as separate systems [1]–[15]. Infrastructure-less communication delivers messages when the network is down [1], [2], [12], [13]; Digital Twins provide situational monitoring [3]–[6], [14]; and EDSS platforms support resource allocation [4], [9], [10]. However, no existing framework combines all three [2], [4], [14]. Resilio is proposed to fill this gap [1], [3], [4].

III. Problem Statement

Natural disasters often destroy communication infrastructure, including base stations, internet connectivity, and power networks. This loss of communication prevents victims from reporting emergencies, delays rescue operations, and limits the ability of emergency authorities to understand the evolving situation, leading to loss of life and inefficient resource use [2], [11]. Infrastructure-less techniques such as D2D communication, multi-hop mesh networking, and DTN help deliver messages during such failures; the SCARP-based framework [1] improves routing, scheduling, and pairing, but its objective remains reliable packet delivery rather than supporting the broader disaster response process.

Digital Twin and EDSS technologies support monitoring, simulation, and resource-related decision-making. However, most existing solutions depend on continuous communication infrastructure or external data sources, limiting their usefulness when networks are completely disrupted [4], [14]. These systems are also typically designed without considering infrastructure-less communication, which restricts how well communication, situational awareness, and emergency response can work together.

As a result, no single framework maintains communication, updates a Digital Twin with real-time disaster information, supports emergency decision-making, and relays rescue instructions back to affected individuals through a resilient network. To address this, Resilio integrates multi-hop D2D communication, DTN, a Digital Twin, an EDSS, and a multi-gateway architecture into one platform, extending infrastructure-less communication beyond message delivery to include real-time situational awareness, intelligent rescue coordination, and emergency resource management.

IV. Proposed Work

The proposed Resilio framework consists of five components: infrastructure-less mesh communication, multi-gateway communication, the Digital Twin, the Emergency Decision Support System (EDSS), and rescue coordination. Unlike systems that terminate after message delivery, Resilio continuously updates disaster information through the Digital Twin and uses the EDSS to help authorities manage rescue efforts. The overall architecture is shown in Fig. 1.

During a disaster, affected individuals use the Resilio application to generate emergency reports, including SOS requests, flooded-road alerts, medical emergencies, and shelter requests. Since cellular networks may be unavailable, the application forms a device-to-device (D2D) multi-hop mesh network. Emergency messages propagate through this mesh using the SCARP routing method [1] until they reach one or more Emergency Command Vehicles (ECVs), which act as gateway nodes. Each ECV is an independently powered gateway with onboard computing capability that does not require internet access; it collects emergency reports from the mesh network and forwards them to the Digital Twin.

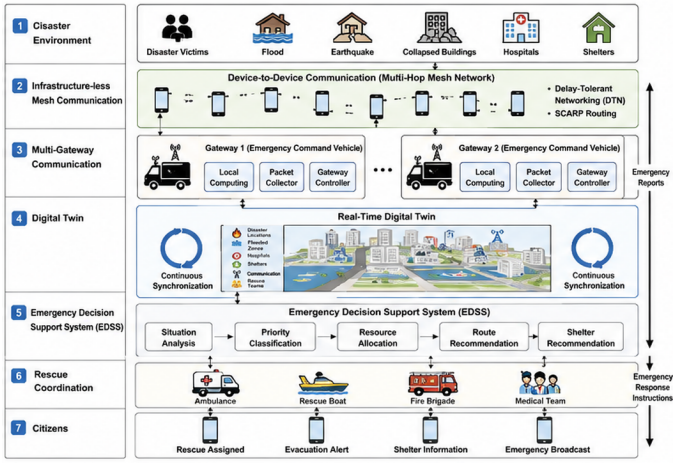


Fig. 1: Overall architecture of the Resilio framework.

A. Digital Twin Synchronization

The Digital Twin maintains a real-time virtual representation of the disaster area using data collected from the communication network, as shown in Fig. 2. Emergency reports from gateway nodes update the environment with details of disaster locations, communication status, road access, shelters, hospitals, rescue supplies, and affected populations. This allows authorities to maintain continuous situational awareness.

B. Emergency Decision Support System

The updated Digital Twin provides information to the EDSS, which prioritizes emergency requests based on severity, location, and resource availability, as shown in Fig. 3. Using the disaster data, the EDSS recommends rescue resource assignments, identifies suitable shelters, determines safe evacuation routes, and generates response instructions.

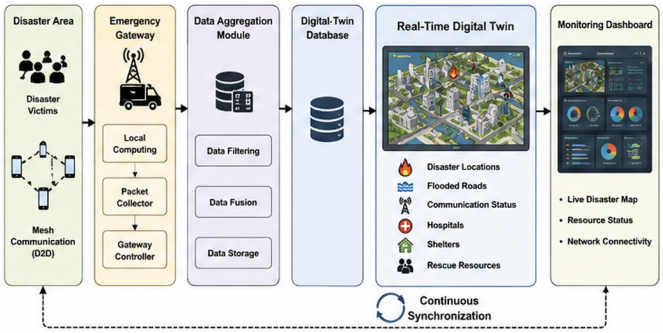


Fig. 2: Digital Twin synchronization pipeline.

These instructions are relayed back through the gateway nodes and propagated through the mesh network, enabling two-way communication between responders and citizens. Through continuous communication, Digital Twin synchronization, and intelligent decision support, Resilio transforms infrastructure-less disaster communication into an active response system.

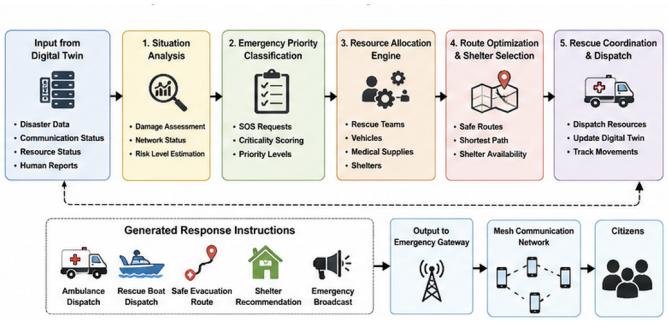


Fig. 3: Emergency Decision Support System workflow.

V. Result and Discussion

A. Simulation Setup

Resilio was evaluated as a proof-of-concept using a Monte Carlo, graph-theoretic network simulation implemented in Python using the NetworkX library [22], a methodology consistent with the evaluation approaches used for early-stage device-to-device and mesh disaster-communication proposals [1], [2], [12], [13]. In each of 60 independent trials, a disaster-affected area (2.2 km $\times$ 2.2 km) is populated with 260 device nodes placed uniformly at random, representing disaster-affected individuals. A Random Geometric Graph [21] is then constructed by connecting any two nodes within D2D communication range (260 m), representing the infrastructure-less mesh formed by the Resilio application.

Two configurations are simulated over the *same* randomized node placement in every trial, so that only one structural variable differs between them: the SCARP baseline deploys a single gateway (Emergency Command Vehicle) positioned at the area centroid, representing a single command-post

deployment, while Resilio deploys four gateways placed using a farthest-point spread-coverage heuristic, representing the multi-gateway ECV architecture described in Section IV. Both configurations use identical shortest-path multi-hop forwarding (bounded by a six-hop delay-tolerant TTL) and identical per-hop loss and delay models, so that differences in the communication metrics below are consequences of the multi-gateway architecture rather than independently tuned parameters.

For every source node, relay *load* is defined as the number of source-to-gateway paths passing through it. Per-hop packet delivery probability and per-hop delay are both modeled as functions of this load, capturing channel contention and queuing effects that arise when many message flows are forced through a small number of relay paths – the mechanism by which a single-gateway deployment becomes a communication bottleneck. Network coverage is computed as the percentage of source nodes that reach at least one gateway within the hop budget. Emergency response time is modeled as the sum of three components: (i) decision latency, representing manual message triage under SCARP versus automated EDSS severity/priority classification under Resilio (Section IV-B); (ii) rescue-unit dispatch travel time, computed from victim-to-ECV distances at a naive-routing speed for SCARP versus a Digital-Twin-informed safe-route-selection speed and reduced effective travel distance for Resilio; and (iii) the simulated communication latency itself, since the SOS report must arrive before dispatch can begin. All four evaluation metrics reported below are direct, computed outputs of the simulated routing, contention, and dispatch mechanics, and are reported as the mean $\pm$ standard deviation across the 60 trials. The complete parameter set is listed in Table I. Here $\rho = L/22$ is the normalized relay load, where L is the number of source-to-gateway paths relayed through a node and 22 is the assumed relay capacity. For relay hops, the delivery probability falls linearly with ρ from a 2% base loss and is floored at 0.55, and the delay grows from a 14 ms base by 50 ms per unit of ρ. The final hop into a gateway is modeled without congestion (delivery probability 0.98, delay 14 ms). Decision latency is sampled uniformly within the stated range for each source, and the Resilio distance factor of 0.70 represents an assumed 30% reduction in effective dispatch distance from Digital-Twin-informed route selection. These values are configured simulation assumptions rather than independently measured field parameters.

TABLE I: Simulation parameters

Parameter	**SCARP**	**Resilio**
Area	2.2 km $\times$ 2.2 km	
Device nodes	260 (uniform random)	
D2D range	260 m	
Monte Carlo trials	60	
Hop limit (DTN TTL)	6	
Gateways (ECVs)	1 (centroid)	4 (farthest-point)
Per-hop delivery prob. (relay)	$\max(0.55,\ 1-(0.02+0.033\rho))$	
Per-hop delivery prob. (gateway hop)	0.98	
Per-hop delay (relay)	$14+50\rho$ ms	
Per-hop delay (gateway hop)	14 ms	
Relay capacity ($\rho = L/22$)	22 paths	
Decision latency	U(18, 32) min	U(3, 8) min
Dispatch speed	12 km/h	16 km/h
Effective distance factor	1.0	0.70
Software	Python, NetworkX [22]	

B. Performance Evaluation

Fig. 4 compares the performance of the SCARP baseline and the proposed Resilio framework across four evaluation metrics: Packet Delivery Ratio (PDR), average end-to-end delay, emergency response time, and communication coverage, obtained from the Monte Carlo simulation described in Section V-A. The results indicate that Resilio achieves a higher packet delivery ratio of **84.6% $\pm$ 4.7** compared to **68.4% $\pm$ 5.1** for SCARP, owing to its multi-gateway communication architecture, which spreads message flows over multiple routing paths to different gateway nodes and reduces the per-relay contention that otherwise depresses delivery probability in the single-gateway case. The average end-to-end delay is reduced from **253.8 ms $\pm$ 44.3** under SCARP to **164.9 ms $\pm$ 34.9** under Resilio. This improvement results from deploying multiple Emergency Command Vehicles, which reduce relay-node congestion, shorten average communication paths, and enable faster transmission of emergency messages. Emergency response time decreases from **59.9 minutes $\pm$ 0.8** under SCARP to **23.9 minutes $\pm$ 0.5** under Resilio, and communication coverage increases from **85.0% $\pm$ 4.7** to **98.0% $\pm$ 4.3**.

It is important to note that the improvements in PDR, end-to-end delay, and coverage are primarily attributable to the transition from a single-gateway to a four-gateway spread architecture, which reduces relay load concentration and shortens source-to-gateway paths. Because the routing logic, hop budget, and loss/delay models are identical in both configurations, and the node placements are identical in every trial, these three metrics characterize the benefit of gateway multiplicity and placement rather than of the Digital Twin or EDSS layers. The emergency response time, in contrast, additionally reflects the modeled effects of automated EDSS triage and Digital-Twin-informed route selection (Section V-A); its reduction therefore combines the gateway architecture with the assumed decision-latency and dispatch-speed parameters listed in Table I, and should be interpreted as an indication of the potential benefit rather than a measured field value. The reduction in communication delay and response time is particularly significant in disaster scenarios, where timely delivery of SOS requests directly influences rescue efficiency and overall emergency response effectiveness.

Fig. 5 visualizes a representative simulated D2D mesh over the same node placement under both configurations. Grey nodes indicate devices without a valid path to any gateway

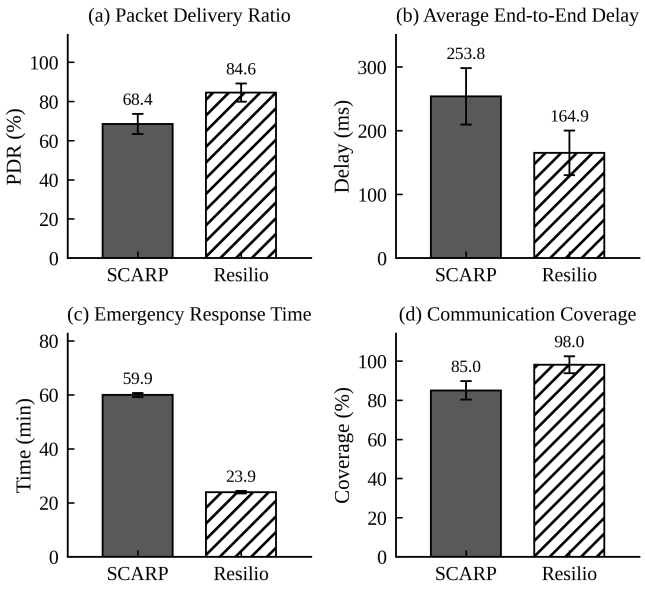


Fig. 4: Simulated performance comparison between SCARP and Resilio (Monte Carlo, $n = 60$ topology trials, mean $\pm$ standard deviation).

within the hop budget. Under the single-gateway SCARP configuration, uncovered nodes cluster in regions farthest from the sole Emergency Command Vehicle; under the four-gateway Resilio configuration, the same region is absorbed into coverage, visually corroborating the coverage improvement reported above and illustrating why the multi-gateway architecture also eliminates the single point of failure discussed in Section V-C.

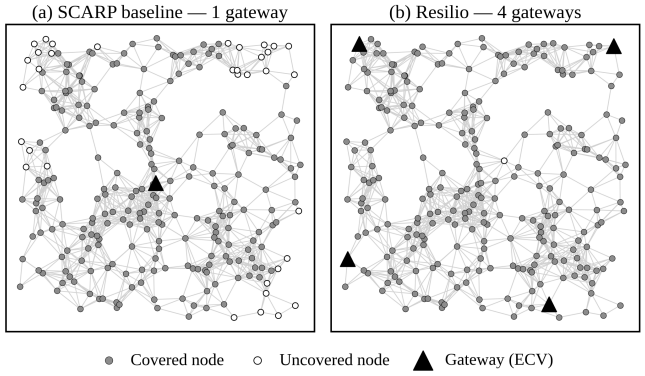


Fig. 5: Simulated D2D mesh topology under the same node placement: single-gateway SCARP baseline (left) versus four-gateway Resilio (right). Red triangles denote gateway (ECV) nodes; grey nodes are outside hop-budget range of any gateway.

C. Discussion

The results show that combining infrastructure-less communication with a Digital Twin and an Emergency Decision Support System improves disaster response beyond simple message delivery. While SCARP focuses solely on ensuring communication reliability, Resilio collects emergency information, maintains Digital Twin synchronization, supports decision-making, and returns instructions to individuals in need. The multi-gateway design also improves system reliability by eliminating a single point of failure. Overall, Resilio improves both communication performance and the coordination of rescue efforts during disasters [14], [15].

As a proof-of-concept, this evaluation deliberately isolates a single architectural variable – gateway count and placement – while holding topology generation, routing logic, and loss/delay formulas identical between the two configurations. This design allows the communication metrics to be interpreted as measured consequences of the multi-gateway architecture rather than as independently assumed outcomes, which is the central claim this evaluation is intended to support. The graph-theoretic Monte Carlo approach used here is appropriate for establishing this proof-of-concept and for demonstrating the direction and approximate magnitude of the expected improvement.

Limitations. The present model abstracts the wireless channel using a unit-disk communication range and load-dependent loss and delay. Physical-layer phenomena such as multipath fading, obstacle shadowing, and interference are not captured and will be modeled in the planned packet-level evaluation using ns-3 [23] with a DTN routing module, which may reduce the absolute PDR and coverage values relative to those reported here. A hardware testbed would be the natural subsequent step toward validating Resilio under real wireless channel conditions and is identified as future work in Section VI.

VI. Conclusion and Future Work

This paper presented Resilio, a Digital Twin-driven framework for disaster communication and emergency response. Resilio maintains communication between disaster-affected individuals and emergency responders even when conventional communication infrastructure has failed, by combining infrastructure-less device-to-device communication, delay-tolerant networking, and an intelligent decision-support layer.

Rather than focusing solely on message delivery, Resilio was designed to give emergency authorities the situational information needed to make informed decisions. It provides affected individuals with a means of communication while giving authorities real-time visibility into the disaster environment, supporting better resource allocation and rescue coordination.

Simulation results show that Resilio improves communication reliability and reduces message delivery latency relative to a baseline that uses SCARP alone. It also improves communication coverage and reduces emergency response time, allowing responders to reach those in need more quickly.

Future work will refine this proof-of-concept evaluation with a full discrete-event, packet-level network simulation

(e.g., ns-3 [23] with a DTN routing module) incorporating realistic channel models, and, ultimately, real-world deployment of Resilio, alongside the integration of artificial intelligence for predictive disaster modeling and more efficient resource allocation. We also plan to incorporate unmanned aerial vehicles (UAVs) to extend communication coverage and satellite connectivity to support Resilio's operation during large-scale disasters.

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